



**Ministry of Higher Education
The Higher Institute of Management
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Job Recommendation System

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Chapter One

The Introduction

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In recent years, the rapid growth of information technology and artificial intelligence has led to significant advancements in how large volumes of textual data are processed and analyzed. With the continuous increase in digital documents such as resumes, job descriptions, and online profiles, the need for systems that can efficiently search, retrieve, and compare textual information has become increasingly important. This challenge has contributed to the development of specialized research fields that focus on managing and understanding text-based data in intelligent ways [1].

One of the core scientific fields addressing this challenge is **Information Retrieval (IR)**. Information Retrieval focuses on the design of systems that can search, organize, and retrieve relevant information from large collections of unstructured or semi-structured data. Instead of manually browsing documents, IR systems aim to automatically identify the most relevant documents based on a user's query. In recruitment systems, Information Retrieval plays a critical role in locating suitable resumes for a given job description or identifying relevant job opportunities for a specific candidate [2].

Closely related to Information Retrieval is the concept of **Text Similarity**, which measures how similar two pieces of text are in terms of meaning and content. Text similarity techniques are essential for comparing resumes with job descriptions, as they allow systems to assess relevance even when different words or expressions are used to describe similar skills or experiences. Rather than relying on exact keyword matches, text similarity enables a deeper comparison based on contextual and semantic relationships between texts [3].

Text Similarity is considered an important application within the broader field of **Natural Language Processing (NLP)**. NLP focuses on enabling computers to understand, analyze, and interpret human language. Through NLP techniques such as text preprocessing, vector representation, and semantic modeling, text similarity algorithms can capture the underlying meaning of documents. This makes them particularly effective for handling real-world recruitment data, where language variability and ambiguity are common [4].

Together, Information Retrieval and NLP-based Text Similarity form the foundation of modern intelligent job-matching and recommendation systems.

These approaches allow systems to rank documents, compute relevance scores, and support decision-making by identifying the best matches between candidates and job requirements. As a result, they have become central research areas in both academic studies and practical recruitment applications [5].

Based on these foundations, this project applies **Information Retrieval (IR)** techniques to retrieve relevant job-related information and leverages widely used **Natural Language Processing (NLP)** methods such as **TF-IDF vectorization** to represent textual data and **cosine similarity** to measure the similarity between a user's CV and a target job title or description. In addition, **machine learning classifiers** are employed to learn from labeled data and support decision-making regarding candidate–job fit. By integrating these IR and NLP approaches into a web-based system, the project demonstrates how theoretical concepts can be effectively applied to build an intelligent job recommendation and CV evaluation system.

1.1 Motivation

This project is motivated by the idea of presenting an innovative concept that has not been widely implemented before. It combines CV analysis and job description matching within a single, simple system that provides a clear and understandable evaluation of candidate suitability. The project aims to offer a direct and practical solution that makes the comparison process more accessible and effective for different users.

Reduce the time and effort required to review CVs manually. Reviewing resumes one by one can be a slow and tiring process, especially when there are many applicants for the same job. This system helps automate the initial evaluation process, allowing faster and more organized screening of candidates.

Its flexibility and wide range of possible users. The system can be useful for students, job seekers, and small organizations, not only large companies. This makes the project practical, scalable, and suitable for different real-world scenarios.

This project help reduce the gap between education and the job market. Many students and fresh graduates are not fully aware of how well their skills match real job requirements. By comparing the CV with the job description, the project helps users better understand the suitability of their skills for available job opportunities

Overall, the motivation for this project comes from the need to simplify and enhance the recruitment process through intelligent automation. By allowing users

to upload their CVs and receive an objective assessment against job requirements, the project demonstrates how AI can support smarter hiring decisions, save time, and improve the experience for both employers and candidates.

1.2 Contributions

- **Automated CV Evaluation:** The system can automatically analyze a candidate's CV against a specific job description, providing an objective assessment of compatibility and reducing reliance on manual review.
- **Improved Matching Accuracy:** By using machine learning and semantic analysis, the project moves beyond keyword-based matching to compare skills, qualifications, and experience in a meaningful way, ensuring more precise candidate-job alignment.
- **Time and Effort Reduction:** Employers and HR teams can save significant time and effort by automating resume screening and initial candidate selection, allowing them to focus on more strategic tasks.
- **Guidance for Job Seekers:** Candidates receive clear feedback on how well their CV fits a particular role, helping them understand strengths and weaknesses in their profiles and enabling more informed and targeted applications.
- **Scalable and Accessible Solution:** The web-based platform integrates AI, machine learning, and modern web technologies to provide a practical, efficient, and scalable system that can be easily accessed by both organizations and job seekers, highlighting the role of intelligent recruitment systems in modern HR practices.

References

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